

ACADEMIC SOFTWARE ENGINEERING PRESENTATION

Computer Store Management System

A Medium-Level Web Application for Hardware Retail Operations, Inventory Control, and Sales Transactions.

 Jakarta EE

 Servlet & JSP

 MVC Architecture

 JDBC & MySQL

 Bootstrap UI

Project Overview & Purpose

Designed to streamline retail store operations by bridging inventory control with client sales management in a unified Java Web platform.

- ✓ **Dual Operational Focus:** Equips store staff with full administrative controls while offering customers an accessible catalog interface.
- ✓ **Medium Complexity Standard:** Moves beyond basic CRUD by incorporating session security, relational integrity, and automated stock processing.
- ✓ **Pure Enterprise Java:** Built with core Jakarta EE components running on Apache Tomcat without heavy external framework bloat.



Problem Statement



Manual Inventory Logs

Paper and spreadsheet-based stock tracking leads to high error rates, misplaced records, and delayed stock updates during peak sales hours.



Stock Discrepancies

Inability to track real-time hardware quantities results in overselling items, unexpected stockouts, and dissatisfied store customers.



Opaque Sales Records

Lack of centralized transaction logs complicates daily auditing, customer order history lookup, and quick store report generation.

Project Objectives



Centralized Data

Establish a unified MySQL database mapping products, brands, categories, sales, and user accounts under strict relational integrity constraints.



Automated Stock Control

Implement automatic stock deduction upon transaction confirmation, enforcing rules to prevent negative product inventory.



Role-Based Access Control

Secure administrative capabilities via HTTP Session management, separating store staff controls from general customer features.

System Scope & Boundaries

✓ Functional Inclusions

- Product, Category & Brand Catalog Management
- Stock tracking with Low-Stock indicator alerts
- Staff sales processing & invoice receipt generation
- Customer product search, filtering, & purchase history
- Session authentication & role-based route guard
- Basic sales & stock summary reports

⊘ Express Exclusions

- Online third-party payment gateways (Credit/PayPal)
- Microservices or complex cloud infrastructure
- AI/ML product recommendation algorithms
- Mobile native apps & Bluetooth/Barcode scanners
- Complex distributed multi-store synchronization

User Roles & Access Permissions

Functional Module	Staff / Administrator	Common User / Customer
Authentication & Profile	Login, Logout, View/Edit Staff Profile	Register, Login, Logout, Edit Profile
Product & Brand Catalog	Full CRUD (Add, Edit, Delete, Categorize)	Read-Only (Browse, Search, Filter)
Inventory & Stock	Adjust Stock, View Low-Stock Alerts	View Stock Status (In-Stock / Out-of-Stock)
Sales & Transactions	Process Sales, Check Out, Print Invoices	Submit Purchase Request / Order
Reports & Dashboard	Access Store Financials & Sales Reports	Access Personal Purchase History Only

Core System Architecture Modules



1. Auth Module

Handles registration, password validation, session creation, and role guards.



2. Product Catalog

Manages hardware specs, pricing, images, categories, and brand tags.



3. Inventory Control

Tracks current quantities, updates stock on sale, and monitors thresholds.



4. Sales Processing

Calculates order subtotals, records sale items, and creates receipts.



5. Search Engine

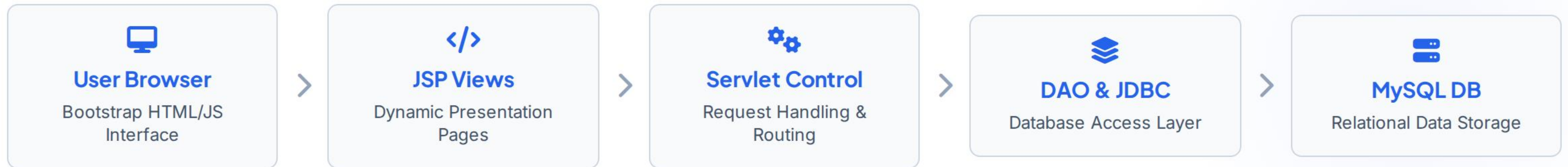
Multi-criteria query filter by keyword, brand, and hardware category.



6. Reports & Metrics

Generates daily/monthly sales reports and stock movement summaries.

System Architecture (MVC Pattern)



-  **Controller (Jakarta Servlets)**: Receives HTTP POST/GET requests, validates user session parameters, invokes service logic, and determines next view navigation.
-  **Model (Java Beans & DAO)**: Encapsulates data objects (Product, Sale, User) and handles SQL execution through JDBC prepared statements.
-  **View (JSP & JSTL)**: Renders data sent from Servlets dynamically using JSTL tags without embedding raw Java scriptlets.

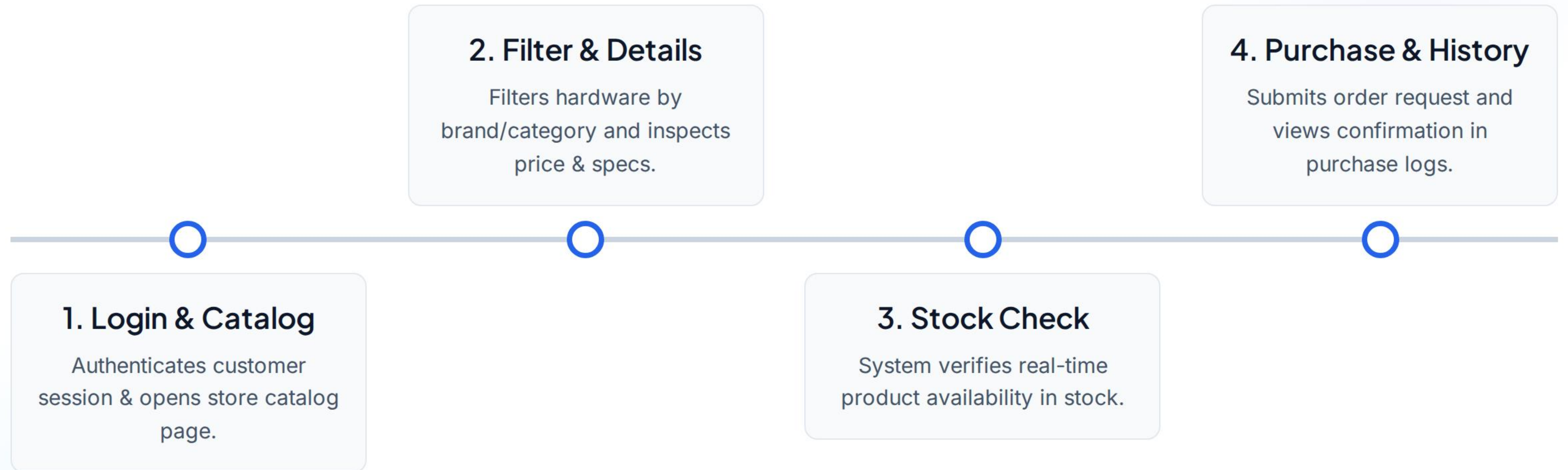
MySQL Relational Database Schema

Table Name	Primary Key (PK)	Foreign Keys (FK)	Key Attributes & Purpose
users	user_id	None	full_name, email, username, password, address
staff	staff_id	None	full_name, email, username, password, role
categories / brands	category_id / brand_id	None	category_name, brand_name, description
products	product_id	category_id, brand_id	product_name, specs, price, status, image
inventory	inventory_id	product_id	quantity, last_updated (1-to-1 with products)
sales	sale_id	user_id, staff_id	sale_date, total_amount, status
sale_items	sale_item_id	sale_id, product_id	quantity, unit_price, subtotal (Junction table)

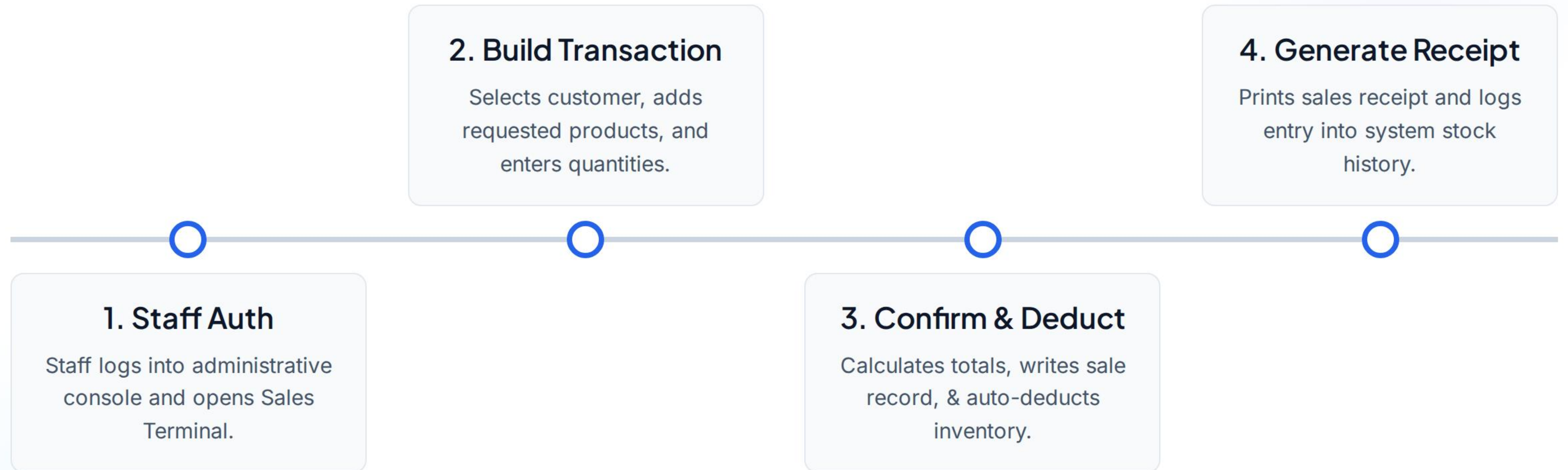
Core Business & Validation Rules

- 🛡️ **Stock Availability Guard:** A transaction can only be processed if `available_quantity >= requested_quantity`. Inventory cannot become negative.
- 🛡️ **Automatic Stock Adjustment:** Product inventory is instantly decremented upon successful transaction confirmation inside a JDBC transaction block.
- 🛡️ **Transaction Integrity:** Every sale record must be associated with a valid customer, authorized staff member, and at least one line item.
- 🛡️ **Data Historical Preservation:** Deleting or soft-deleting a product does not remove or alter past historical sales transactions or invoice logs.
- 🛡️ **Data Privacy & Security:** Customers are restricted to accessing their own profile and order history via HTTP Session guards.

Customer Experience Workflow



Staff Sales Processing Workflow



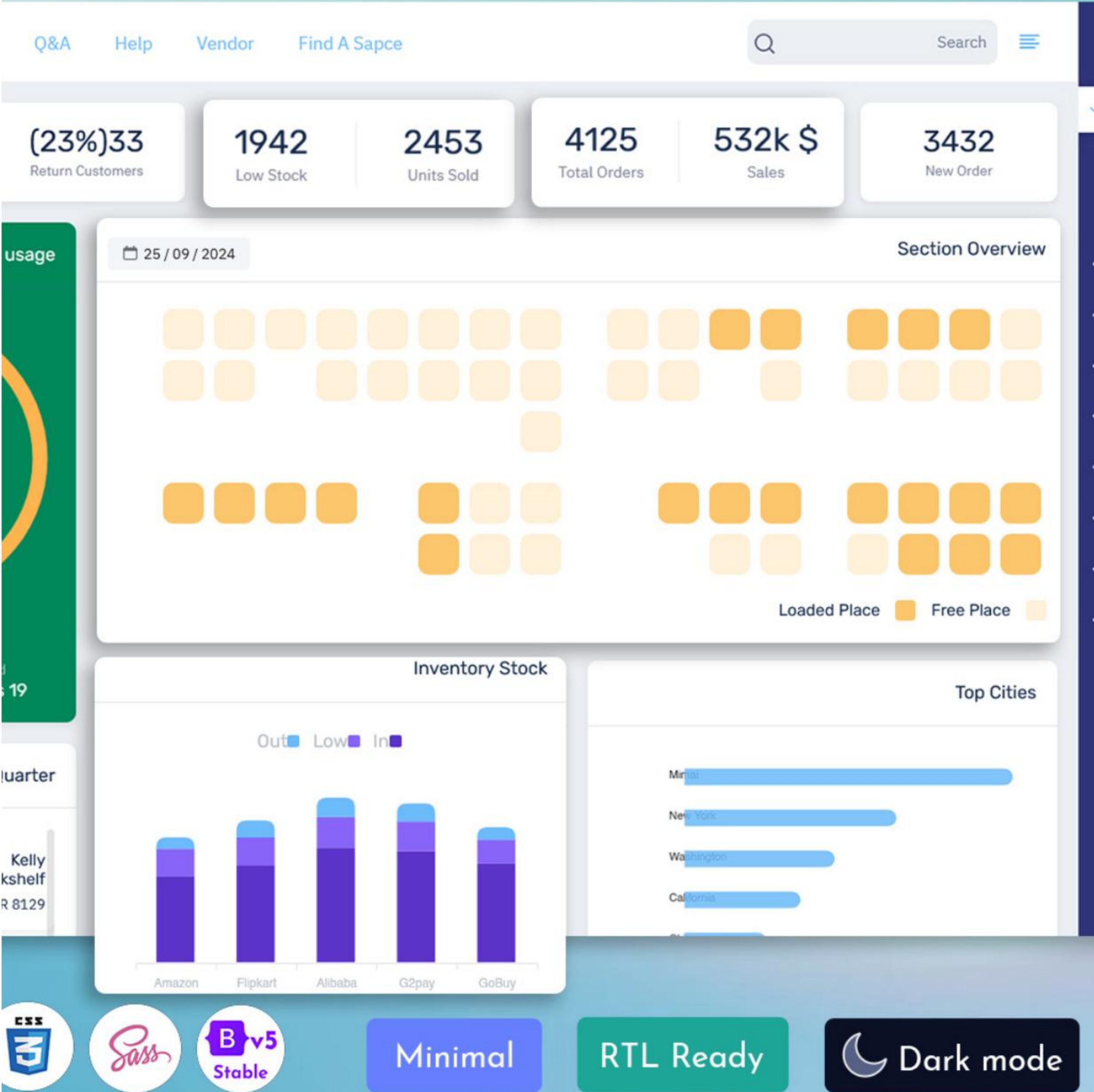
System UI & Dashboard Design

Responsive Bootstrap Interface

The front-end utilizes clean Bootstrap layout components to provide intuitive navigation for both technical store staff and end-user customers.

-  **Admin Dashboard:** Real-time KPI summary widgets displaying total products, low-stock warnings, and revenue metrics.
-  **Customer Portal:** Card-based product grids with brand badges, stock pill indicators, and quick specification popups.

House Responsive Bootstrap 5 Admin Dash



Complete Technology Stack

Layer / Technology	Specific Component	Role in System
User Interface	HTML5, CSS3, JS, Bootstrap 5	Provides responsive view pages, forms, and dynamic tables.
Server-Side Platform	Jakarta EE (Servlet 5.0 / JSP)	Handles HTTP requests, session management, and routing.
Architecture Pattern	MVC (Model-View-Controller)	Decouples UI views from business logic and database access.
Database Access	JDBC (Java Database Connectivity)	Executes parameterized SQL queries & manages transactions.
Database Engine	MySQL RDBMS	Stores relational user, product, inventory, and sales data.
Web Server & Environment	Apache Tomcat 10 / Eclipse IDE	Runs Servlet container and hosts application deployment.

Benefits, Limitations & Future Scope

System Benefits

- Eliminates manual stock tracking errors with instant database updates.
- Enforces role authorization and strict stock availability rules.
- Demonstrates enterprise Jakarta EE concepts in a clean academic architecture.

Future Enhancements

- Barcode / QR scanner hardware integration for rapid checkout.
- Email notifications for low stock alerts & digital receipts.
- RESTful API endpoints for external mobile application connectivity.

Image Sources



<https://epcusa.com/wp-content/uploads/sites/9/2021/02/Retail-Glory-Shot-e1783536932903.jpg>

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